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Direct estimation of entropy and revisions to the national income accounts

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Abstract. This study reports an analysis of the reduction in uncertainty, or information gain, contained in data which is revised subsequent to the publication of preliminary estimates. A central concept in the analysis is that of the entropy of a distribution. We develop existing methods to allow for non-normality and directly estimate the entropy of the appropriately defined conditional distributions. We show that a failure to account for non-normality would give rise to misleading conclusions.

1 Introduction

Much non-experimental data, such as that used by economists and other social scientists, is first released by statistical agencies in provisional form. It is then updated—or revised—as further information becomes available. Examples of such data include most of the main series which constitute the national income accounts and time series of key monetary aggregates. The process of revision implies that different *vintages* of data are available for a particular variable for a given time period. The existence of more than one vintage of data raises the question of which vintage should be used in empirical work. While the general answer to this question will often depend upon the particular use to which the data is going to be put, one important consideration will be the extent to which the process of revision leads to a reduction in uncertainty as to the final vintage of data. If, for example, the second and subsequent vintages of data lead to no, or little, reduction in uncertainty over the first vintage, as to the final outcome then the first vintage may well prove adequate for the purpose at hand.

Such a consideration is important for another reason. Revising initial estimates takes time; for example, the final outcome of some national income aggregates is not known until (at least) 6 years after publication of the initial estimate. Hence, if the final vintage is to be used 6 years of data are necessarily curtailed from the sample size. Given that in both statistics and econometrics large samples are often thought desirable, leading to benefits in the properties of estimators, there is the very real practical consideration of whether ‘it will do’ to use preliminary data. Consider the situation in which subsequent vintages of data lead to substantial reductions in uncertainty as to the final outcome. The applied researcher then faces a trade-off between sample size and uncertainty reduction. In practice though the, albeit, casual impression we have is that insufficient attention is paid to the characteristics of different vintages of data despite evidence that data vintage matters. For example, Maravall & Pierce (1983) show, in the context of the U.S. monetary authorities reacting to growth rates of M1 outside specified target bands, that using later vintages of data reverses important policy conclusions based on the initial vintage.

Using a data set, made available to us by the United Kingdom Central Statistics Office, comprising the seasonally adjusted constant price expenditure components and current price income components of gross domestic product (GDP), our focus is to assess the reduction in uncertainty afforded by later vintages of data compared with the earlier vintages. The data set is an important one because it comprises much studied series on, for example, consumption and investment expenditure, exports and imports of goods and

services in the constant price series, and income from employment and corporate profits in the current price series. It is also, to our knowledge, the most extensive data set of revisions, comprising 25 vintages of observations on the same series, to have been made available.

We do not know of any studies which have addressed the question of uncertainty reduction in the context of data vintages, although others have considered this area—see, for example, Holden & Peel (1982). Some related work was undertaken by Theil (1967) and Theil & Scholes (1967) who assessed whether in the case of multi-stage forecasting, in which an initial forecast would be revised several times, uncertainty reductions resulted from moving to later stages in the forecasting process. However, although important in setting out the framework for an analysis of uncertainty reduction, Theil's framework needs to be extended in two important aspects before it can be more generally applicable. This we do.

As in Theil (1967) our central concept is that of the entropy of a distribution defined as

$$H = - \int_{-\infty}^{\infty} f(y) \ln f(y) dy \quad (1)$$

for a continuous distribution with density function $f(y)$. The base of logarithms is immaterial—see Kullback (1968, p. 5)—and we use Napierian logarithms throughout. The concept of entropy, or disorder, is a measure of uncertainty originating in thermodynamics and statistical mechanics, and is closely related to the concept of information in communication theory and mathematical statistics—see, for example, the seminal paper by Shannon (1948) and also Shannon & Weaver (1949), Kolmogorov (1956) and Mandelbrot (1956). For an application in the context of data revisions consider the distributions obtained from observations on a particular variable at the r -th and $(r+1)$ -th stage of the revisions process. These are, effectively, distributions conditional on the vintage of data. If the difference between the entropies of these distributions is positive there is a reduction of uncertainty attributable to the later vintage of data compared to the earlier vintage. A priori, though, we could not rule out an increase in uncertainty. Whether or not this occurs in practice is of importance to data users and data producers.

In Theil's original application it was assumed, throughout, that $f(y)$ was the density function associated with the normal distribution. This was a useful assumption since in that case $H = 0.5 \ln 2\pi e \sigma^2$, where σ^2 is the variance and all that is then required is an estimate of σ^2 . It was also implicitly assumed that the means of the successive distributions, conditioned on the different stages in the forecasting process, were constant. Neither of these assumptions is tenable for data revisions—or multi-stage forecasting. First, many of the conditional distributions exhibit a marked departure from normality. Second, even if normality is a reasonable assumption that of constant means is not, as there is a tendency for data revisions to be positive and decline with later vintages of data. These problems can be overcome by *direct* estimation of the entropy of the appropriately defined conditional distributions, using a development due to Vasicek (1976). For the case where the distribution is normal, or statistically indistinguishable from the normal, we derive an expression for the reduction in uncertainty without assuming constancy of the means.

The organisation of this paper is as follows. The next section contains a review of concepts and sets the analysis in the context of data revisions. Here we show how the framework for an evaluation of uncertainty reduction, due primarily to Theil, can be developed to make it more generally applicable. The third section reports the empirical results of our analysis applied to the current price income components and constant price expenditure components of GDP. Here we show that it is necessary to take account of non-normality and non-constant means in order to obtain an accurate indication of the reduction in entropy available from using later rather than earlier vintages of data. The fourth section contains some concluding remarks.

2 A review of concepts

We first establish some notation and set the context of the analysis. Let y_{itv} denote the v -th vintage of the i -th variable for the t -th observation in a sample of size T . There are, at maximum, m vintages and j variables. Different vintages of the data arise because new information becomes available, after the first vintage of data has been published, leading to revisions, or refinements of the preliminary estimates. It might be, for example, that the initial data is based on a sample of all of the returning organisations, later vintages correct for late reporting in the initial sample and for processing errors, and more substantial revisions arise when all organisations have reported. The initial and preliminary vintages could also be regarded as forecasts, at different stages, of the final outcome, y_{itm} , the different stages being represented by the $v = 1, \dots, (m-1)$ vintages.

2.1 An analysis assuming normality

Given different vintages of data on the same variable the question then arises as to how to assess the accuracy of the preliminary vintages of data with respect to the final outcome. An analogous problem was considered by Theil (1967) and Theil & Scholes (1967), who were concerned with measuring the inaccuracy of multi-stage forecasts with respect to the eventual realisation. For simplicity of exposition assume, initially, that the distribution of the (cumulative) revision defined as $y_{itv} - y_{itm}$ is normal, with variance σ_{itv}^2 ; then the conditional distribution of y_{itm} given y_{itv} is also normal with variance σ_{itv}^2 . We will use the entropy of this conditional distribution as a measure of uncertainty of the final outcome. Given normality we just require an estimate of σ_{itv}^2 in order to calculate the entropy.

There are, however, $(j \cdot T)m$ combinations of the i, t, v indices resulting in non-zero variances. Thus, for our data sets there are 7675 and 5475 combinations for the expenditure and income components of GDP, respectively. A simplifying assumption as to the structure of these variances is, therefore, required. Following Theil, we will use a multiplicative decomposition of the variance, σ_{itv}^2 . In our context this decomposition involves three factors: (i) the 'vintage' effect, that is keeping the variable and quarter indices constant we anticipate a reduction in the variance as we move from an earlier to a later vintage of data; (ii) the 'variable' effect, this reflects the likely situation that the different variables in our data set, for example, income from employment, corporate profits, are subject to different particularities of the revisions process (some might be particularly subject to late reporting or otherwise incomplete initial reports); and (iii) the 'quarter' effect, reflecting the possibility that some quarters or years may have characteristics either associated with the revisions process in general (for example, suppose data reporting and processing is more difficult in Q3 of each year because that in the peak time for holidays) or with particular quarters (for example, periods of rapid change, as in the upturns of the economic cycle, may cause problems with data collection).

More formally, the variance we are considering is defined as follows,

$$\text{var}(y_{itm}|y_{itv}) = E[\{(y_{itm} - y_{itv})|y_{itv}\} - E\{(y_{itm} - y_{itv})|y_{itv}\}]^2 \quad (2)$$

$$= E[(y_{itm} - y_{itv})^2|y_{itv}] - E[(y_{itm} - y_{itv})|y_{itv}]^2 \quad (3)$$

If the average revision is zero (3) reduces to the first term on the right hand side, which is the second moment of y_{itm} around y_{itv} given y_{itv} , or the mean square revision. This is then decomposed multiplicatively in the following manner,

$$E[(y_{itm} - y_{itv})^2|y_{itv}] = A_i^2 B_t^2 C_v^2 \quad \text{for } \begin{matrix} i = 1, \dots, j \\ t = 1, \dots, 44 \\ v = 1, \dots, 24 \end{matrix} \quad (4)$$

Where $j=5$ and 7 for the income and expenditure components of GDP, respectively. For each variable there are 44 'quarter' coefficients, and 24 'vintage' coefficients, and there is then one further coefficient for each variable. Hence, a comparison amongst variables concentrates on the differences in the A_i coefficients; a comparison of different quarters looks at variation in the B_t coefficients. In the special case of normality and constant means, of the conditional distributions, the reduction in uncertainty comparing vintages v and $v+\delta$ is a simple function of the C_v and $C_{v+\delta}$ coefficients. We examine this special case next as a prelude to a more general method.

Given normality and zero means the difference between entropies of the conditional disturbances of y_{itm} given y_{itv} and $y_{itv+\delta}$, respectively is,

$$H(y_{itm}|y_{itv}) - H(y_{itm}|y_{itv+\delta}) = \ln(C_v/C_{v+\delta}) \quad (5)$$

for δ , an integer, >0 , and where we have used the decomposition given in (4) above. In this case the reduction in uncertainty is a simple function of the 'vintage' coefficients, with an information *gain*—or uncertainty reduction—if $C_{v+\delta} < C_v$, and an information *loss*—or uncertainty increase—if $C_{v+\delta} > C_v$.

There are two consequences of incorrectly assuming zero means for the conditional distributions, one of which was noted by Theil (1967). However, the other and more serious consequence needs also to be taken into account. Note, first that it is a general finding in our data sets that the initial vintages of data are subject to an upward revision implying that $E[(y_{itm} - y_{itv})|y_{itv}]$ is >0 ; further, these conditional means are not constant, getting closer to zero as v approaches the final vintage. This implies that the mean square revision overestimates the variance by a factor which depends upon the mean revision. The extent of the problem can be seen on defining the following standardised revision,

$$\epsilon_{itv} = (y_{itv} - y_{itm})/A_i B_t C_v \quad (6)$$

If we find that $E\epsilon_{itv} = \epsilon \neq 0$, then the ratio of the conditional variance (2) to the mean square revision (4) is $(1 - \epsilon^2)$ —see also Theil (1967, p. 405). Given that ϵ is in standardised units the degree of over-estimation need not be large; for example, if $\epsilon = -1/4$, then $(1 - \epsilon^2) = 15/16$. In practice, $E\epsilon_{itv}$ is unlikely to be constant for each i, t, v combination; this causes problems not so much for the overall decomposition (4), provided the conditional means are not large, but for expression (5) for the difference between entropies, which is based on constant means (zero being a special case). Denoting the mean of $y_{itm} - y_{itv}$, conditional on the vintage v , as μ_v expression (5) becomes

$$H(y_{itm}|y_{itv}) - H(y_{itm}|y_{itv+\delta}) = 0.5 \ln[(C_v^2 - \mu_v^2)/(C_{v+\delta}^2 - \mu_{v+\delta}^2)] \quad (7)$$

Thus (7) is the appropriate expression for measuring the information gain or uncertainty reduction comparing vintages v and $v+\delta$. It reduces to (5) if $\mu_v = \mu_{v+\delta}$, which is, of course, satisfied if the conditional means are zero. Before considering how to extend the analysis to allow for non-normality we need to take into account the different scale of the variables we are considering. Note that the mean square revision, (4), is not normalised (for the scale of variables). For example, an upward trend in the 'quarter' coefficients for a particular variable may arise because over time the earlier vintages have deteriorated in quality as indicators of the final vintage *or* because there has been very much more change in the variable. This is likely to be particularly relevant to the data on the *current* price income components of GDP which are, in general, strongly trended. Following Theil we assume that y_{itm} , the final vintage of variable i in period t , is the outcome of a stochastic process which satisfies,

$$E y_{itm}^2 = A_i'^2 B_t'^2 \quad \begin{matrix} i=1, \dots, j \\ t=1, \dots, 44 \end{matrix} \quad (8)$$

Where A'_i measures the degree of change in the i -th variable and B'_t the degree of change in the t -th quarter. A normalisation, or scaling, of the mean square revision is then given by the ratio of (4) to (8), ie,

$$E[(y_{iim} - y_{iiv})^2 | y_{iiv}] / Ey_{iim}^2 = (A_i / A'_i)^2 (B_t / B'_t)^2 C_v^2 \quad (9)$$

2.2 Relaxing the normality assumption

An important qualification to the method so far outlined is the assumption that the appropriately defined conditional distributions are normal. As we report in the next section this only, generally, comes close to being a reasonable assumption when all of the component variables are pooled; but pooling is only of limited interest as we are not then able to assess whether there are differences amongst variables in the pattern of information gains. The normality assumption can be dropped if we estimate entropy directly from the empirical conditional distributions. We can then compare the estimated entropies of two distributions, one conditional on variable i and vintage v and the other on variable i and vintage $v + \delta$. The difference between entropies is then due to the change in vintage.

The method we adopt for estimating entropy is a modification of that due to Vasicek (1976). Denote the cumulative density function $F(y) = p$, then by a change of variables from y to p , Vasicek shows that the entropy of a distribution, (1), can be expressed as,

$$H = \int_0^1 \ln [dF^{-1}(p)/dp] dp \quad (10)$$

A consistent estimator of H , denoted H_{mn} , is then obtained by replacing the distribution function $F(p)$ by the empirical distribution function, and using the difference operator in place of the differential operator. Let $y_1, y_2, \dots, y_n$ be a sample from the distribution and let $y_{(1)} \leq y_{(2)} \leq \dots \leq y_{(n)}$ be the associated order statistics, then the derivative of $F^{-1}(p)$ is estimated as follows:

$$\begin{aligned} & (y_{(i+m)} - y_{(i-m)})n/2m \text{ for } (i-1)/n < p \leq i/n; i = m+1, \dots, n-m; \text{ say } i \in I_1 \\ & (y_{(i+m)} - y_{(1)})n/(i+m-1) \quad \text{when } p \leq m/n; \text{ say } i \in I_2 \\ & (y_{(n)} - y_{(i-m)})n/(n-i+m) \quad \text{when } p > (n-m)/n; \text{ say } i \in I_3 \end{aligned}$$

Where m is a positive integer smaller than $n/2$.

The estimator H_{mn} of H is,

$$\begin{aligned} H_{mn} = n^{-1} & \left[\sum_{i \in I_1} \ln(y_{(i+m)} - y_{(i-m)})/2m \right. \\ & \left. + \sum_{i \in I_2} \ln(y_{(i+m)} - y_{(1)})/(i+m-1) + \sum_{i \in I_3} \ln(y_{(n)} - y_{(i-m)})/(n-i+m) \right] \quad (11) \end{aligned}$$

Vasicek (1976) also derives a test for normality based on (11). It is well known in information theory that of all distributions which possess a density function, $f(y)$, and have a given variance σ^2 , and a range of variation $(-\infty, \infty)$, the normal distribution has the largest entropy equal to $0.5 \ln 2\pi e \sigma^2$. A test statistic for normality is,

$$K_{mn} = \exp(H_{mn})/s \quad (12)$$

with $s^2 = \Sigma(y_i - \bar{y})^2/n$, $0 \leq K_{mn} < (2\pi e)^{1/2} = 4.133\dots$, and critical region $K_{mn} \leq K^*$. 5% critical points have been tabulated by Vasicek for $n \leq 50$, and $m = 1, 2, \dots, 5$, who also shows that the 'K' statistic has good power characteristics against the exponential and Cauchy distributions and performs respectably for the Gamma, Uniform and Beta distributions. Comparing the 'K' statistic with 6 other tests in a Monte Carlo Study,

Vasicek found it had the highest power against at least three of these five alternative distributions.

Using these developments we can estimate the entropy of the relevant conditional distributions and, by taking the difference in entropy for two successive conditional distributions, obtain an estimate of the information gain. To the extent that the empirical distributions are not normal, the estimated information gain will differ from that assuming normality. If normality is a tenable assumption, which could be established by calculating the 'K' statistic, we can either use (7) above to measure the information gain or *equivalently* calculate the entropy on the assumption that the conditional distributions are normal using the estimated variance from the empirical distributions.

3 Empirical results

3.1 Variable and quarter effects

The results we report in this section are for the five current price income components of GDP (GDPI) and seven constant price expenditure components of GDP (GDPE). For each of these sets of time series made available to us by the CSO we have 44 observations on 25 vintages of data, covering the period 1970Q1 to 1980Q4. Our analysis of these results focuses on three central concerns: (i) the 'variable' coefficients which indicate the differences in mean square revision amongst the respective sets of GDP components; (ii) the 'quarter' coefficients which indicate variations in mean square revisions over time; and (iii) the 'gain' in going from one vintage of data to a later vintage. It is this latter concern which is, perhaps, the primary one for if all of the gain is in the early vintages the later vintages can, effectively, be ignored. In connection with establishing the gain we are also interested in the practical importance of the development of Theil's methodology to (a) correct his expression for the gain *if* the empirical frequency distribution is (close to) normality, and (b) estimate the entropy of these distributions, which becomes important when normality is *not* a tenable assumption. These developments are likely to be of importance because whilst, like Theil, we present the results of pooling all of the variables (for each set of components in our case), we go further and consider each of the variables separately. One result of pooling the variables is to increase the possibility that the overall empirical frequency distribution is close to normality, in contrast to the conditional (on vintage) empirical frequency distributions for *each* variable which are in many cases far from normal.

Table 1 reports the variable coefficients, A_i , and the normalised variable coefficients, A_i/A'_i . It is these latter coefficients which are of greater interest as they adjust for the scale of the variable being considered and, hence, are comparable across variables. Thus, whilst the A_i coefficients indicate that income from employment has the second largest 'variable' coefficient amongst the GDPI components, it has the smallest normalised coefficient. In this set of components the largest 'variable' coefficient is for corporate profits, closely followed by stock appreciation. Turning to the GDPE components, note that stockbuilding has by far the largest normalised 'variable' coefficient; this variable apart, the remaining coefficients are of a similar order of magnitude with the exception of investment, which has a normalised coefficient about twice that of the best variable.

The 'quarter' coefficients obtained by pooling all of the components in GDPI and GDPE, respectively, are shown in Table 2. Considering the GDPI components first, we note a tendency for the B_t coefficients to increase over time and a seasonal variation, with the coefficients for quarter 1 larger, on average, than for the other quarters. To see if these patterns are just due to the changing scale of the variables over time we also present the normalised coefficients, B_t/B'_t . These show a much more constant pattern over time with,

Table 1. The 'variable' coefficients

GDPI components	A_i	A_i/A'_i	Ranking	GDPE components	A_i	A_i/A'_i	Ranking
Income from employment (YE)	358.9	0.022	1	Consumption (C)	247.3	0.027	3 =
Other income (OY)	299.8	0.077	3	Investment (I)	120.4	0.048	5
Gross trading profits of public corporations (PCP)	49.2	0.063	2	Government Exp., (G)	61.6	0.023	1
Stock appreciation (SA)	215.4	0.218	4	Exports (X)	88.7	0.024	2
Gross trading profits of companies (CP)	841.1	0.254	5	Imports (M)	93.4	0.027	3 =
				Factor Cost Adj., (FCA).	75.1	0.034	4
				Stockbuilding (SB)	104.1	0.611	6

Table 2. 'Quarter' coefficients (annual average for pooled components)

GDPI components										
	Q1		Q2		Q3		Q4		Average	
	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t		
1970	0.417	0.982	0.478	1.143	0.484	1.146	0.547	1.232	0.482	1.126
1971	0.552	1.232	0.584	1.182	0.447	0.944	0.515	1.089	0.525	1.112
1972	0.557	1.199	0.608	1.163	0.632	1.121	0.613	1.046	0.603	1.132
1973	0.672	1.096	0.796	1.211	0.644	0.854	0.643	0.790	0.689	0.988
1974	0.888	0.793	1.339	1.343	0.932	0.981	0.988	0.973	1.037	1.023
1975	1.710	1.548	1.017	0.997	1.611	1.534	0.988	0.853	1.332	1.233
1976	1.664	1.455	1.260	0.947	1.443	1.033	0.936	0.631	1.326	1.017
1977	1.380	0.913	1.022	0.733	1.230	0.835	1.226	0.910	1.215	0.848
1978	1.637	1.178	1.519	0.968	1.446	0.908	1.481	0.907	1.521	0.990
1979	1.940	1.094	1.644	0.811	2.066	0.993	1.635	0.785	1.821	0.921
1980	2.022	0.879	1.855	0.930	1.341	0.686	1.663	0.845	1.720	0.835
Average	1.222	1.124	1.102	1.039	1.116	1.003	1.021	0.915	1.115	1.020

GDPI components										
	Q1		Q2		Q3		Q4		Average	
	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t		
1970	0.690	0.903	0.632	0.725	0.668	0.775	0.573	0.703	0.641	0.777
1971	0.949	1.129	1.479	1.778	1.161	1.412	1.059	1.275	1.162	1.399
1972	0.951	1.138	1.488	1.741	0.864	1.030	1.036	1.162	1.085	1.268
1973	1.243	0.988	1.592	1.234	1.501	1.260	1.266	1.190	1.401	1.168
1974	1.324	1.436	1.023	0.875	0.837	0.694	0.937	1.013	1.030	1.005
1975	1.409	1.295	0.850	0.836	1.069	1.101	0.944	1.020	1.068	1.063
1976	1.080	1.169	0.905	0.959	0.751	0.771	0.873	0.795	0.902	0.924
1977	0.935	0.871	0.867	0.803	0.725	0.760	0.928	0.889	0.864	0.831
1978	1.116	1.101	1.331	1.228	0.836	0.807	0.874	0.843	1.039	0.995
1979	1.288	1.137	1.163	1.074	0.972	0.810	0.918	0.846	1.085	0.967
1980	1.544	1.406	0.735	0.706	1.104	0.969	0.892	0.695	1.069	0.944
Average	1.139	1.143	1.097	1.087	0.953	0.944	0.936	0.948	1.031	1.031

broadly, a decline in the mean square revision after 1975. However, the seasonal pattern remains in the normalised coefficients, with the average for quarter 1 some 12% higher than the average of the other quarters.

As the GDPE components are in constant prices we expect less of a scale effect compared to the GDPI components, and this is borne out in Table 2 where the normalisation has much less of an impact. As with the GDPI components there is a downward tendency in the normalised coefficients, indicating a reduction in the mean

Table 3. 'Quarter' coefficients for individual variables (average by year and quarter)

GDPI components											
YE			OY			PCP			SA		
B_t	B_t/B'_t	B_t	B_t	B_t/B'_t	B_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t
1970	0.496	1.018	0.750	1.661	0.732	1.562	0.441	1.375	0.378	0.860	
1971	0.487	0.911	0.590	1.157	0.402	0.860	0.582	2.032	0.625	1.223	
1972	0.891	1.480	0.631	1.048	0.663	1.214	0.438	1.309	0.430	0.749	
1973	0.670	0.977	0.569	0.747	0.614	0.921	0.956	1.438	0.536	0.736	
1974	1.065	1.263	0.713	0.811	0.567	0.678	1.918	1.143	0.909	1.123	
1975	1.038	0.974	1.259	1.257	1.758	1.825	1.299	0.813	1.528	1.649	
1976	1.232	1.002	1.515	1.267	1.259	0.894	1.475	0.931	1.404	1.211	
1977	0.767	0.559	0.858	0.637	0.970	0.583	1.704	1.188	2.311	1.489	
1978	2.042	1.307	1.611	0.996	0.880	0.529	0.906	0.872	2.021	1.162	
1979	2.255	1.222	2.040	1.086	4.478	2.477	1.836	0.731	1.679	0.771	
1980	2.338	1.084	2.062	0.946	4.459	2.284	1.762	1.047	1.545	0.707	
Q1	1.239	1.085	1.198	1.201	1.639	1.420	1.494	1.424	1.287	1.107	
Q2	1.230	1.142	1.144	1.052	1.318	1.129	1.173	1.053	1.143	1.014	
Q3	1.117	1.010	1.051	0.957	2.028	1.551	1.116	1.045	1.216	1.055	
Q4	1.243	1.053	1.188	1.014	1.120	0.928	1.060	1.162	1.214	1.071	

Table 3 continued

GDPE components													
C		I		G		X		M		FCA		SB	
B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t	B_t	B_t/B'_t
1970	0.643	0.732	1.407	1.460	0.441	0.508	0.470	0.616	0.321	0.410	0.480	0.566	0.871
1971	0.922	1.020	1.606	1.622	1.185	1.339	2.267	2.716	2.202	2.615	0.460	0.524	0.492
1972	0.883	0.917	1.364	1.389	1.111	1.215	1.513	1.777	2.130	2.284	0.841	0.891	0.973
1973	1.477	1.457	1.583	1.525	0.932	0.963	1.377	1.451	1.989	1.914	1.228	1.199	2.284
1974	0.998	1.000	1.645	1.633	0.962	0.979	1.148	1.146	1.095	1.040	0.776	0.799	1.548
1975	1.362	1.367	0.804	0.804	1.649	1.583	1.092	1.111	0.859	0.882	1.061	1.089	1.185
1976	0.945	0.949	0.671	0.663	1.062	1.012	1.177	1.105	1.034	1.032	1.046	1.054	1.065
1977	1.121	1.123	0.612	0.621	1.119	1.077	0.855	0.751	1.066	1.038	1.325	1.329	0.888
1978	1.468	1.385	0.525	0.514	1.082	1.010	0.869	0.752	1.269	1.172	1.680	1.512	1.240
1979	1.198	1.078	0.971	0.942	1.441	1.312	0.897	0.742	0.398	0.337	2.044	1.761	1.250
1980	0.745	0.669	0.993	1.012	1.109	0.996	0.750	0.621	1.483	1.287	2.970	2.655	0.934
Q1	0.995	0.992	1.232	1.231	1.254	1.240	1.145	1.194	1.529	1.518	1.584	1.520	1.278
Q2	1.311	1.305	1.093	1.091	1.098	1.100	1.216	1.268	1.376	1.423	1.296	1.236	1.202
Q3	1.034	1.025	1.122	1.125	1.112	1.110	0.962	0.980	0.964	0.980	1.016	0.994	1.187
Q4	0.937	0.932	0.981	0.984	0.993	0.911	1.191	1.209	1.165	1.174	1.162	1.108	0.962

Note: see Table for definitions of the GDPI and GDPE components.

Table 4. Entropy and gain—the conditional distributions

Entropy			Entropy		
Normal	Direct	K-test ^a	Normal	Direct	K-test ^a
GDPI (Pooled Components)					
<i>v</i> = 1	1.248	1.220	1.315	1.227	3.786
5	0.833	0.768	1.056	0.965	3.771
9	0.616	0.559	0.813	0.732	3.814
13	0.369	0.384	0.542	0.479	3.883
17	0.029	0.083	0.339	0.268	3.851
21	-0.222	-0.503	-0.020	-0.147	3.643
Income from employment (YE)					
<i>v</i> = 1	0.982	0.815	0.774	0.610	3.508
5	0.880	0.558	0.325	-0.013	2.945
9	0.741	0.148	-0.029	-0.208	3.458
13	0.433	0.107	-0.263	-0.439	3.468
17	0.149	-0.279	-0.238	-0.433	3.398
21	0.112	-0.116	-0.491	-0.705	3.336
Other income (OY)					
<i>v</i> = 1	0.905	0.696	1.022	0.803	3.319
5	0.563	0.709	0.682	0.406	3.135
9	0.328	0.025	0.567	0.402	3.507
13	0.013	-0.259	0.285	0.297	3.155
17	0.072	-0.185	-0.038	-0.452	2.733
21	-0.158	-0.406	-0.387	-0.583	3.399
Public corporation profits (PCP)					
<i>v</i> = 1	1.316	1.037	1.416	1.282	3.615
5	0.472	0.127	1.512	1.221	3.088
9	0.183	0.222	1.504	1.120	2.846
13	-0.399	-0.749	1.488	1.170	3.008
17	-2.093	-2.293	1.392	1.209	3.440
21	-3.089	-3.367	0.782	0.573	3.352
GDPE (pooled components)					
<i>v</i> = 1	—	—	—	—	—
5	0.414	0.452	1.056	0.965	0.259
9	0.218	0.209	0.813	0.732	0.243
13	0.247	0.175	0.542	0.479	0.271
17	0.339	0.351	0.339	0.268	0.203
21	0.252	0.536	-0.020	-0.147	0.360
Consumption (C)					
<i>v</i> = 1	—	—	0.774	0.610	—
5	0.101	0.257	0.325	-0.013	0.448
9	0.140	0.410	-0.029	-0.208	0.354
13	0.307	0.041	-0.263	-0.439	0.234
17	0.284	0.387	-0.238	-0.433	-0.025
21	0.036	-0.164	-0.491	-0.705	0.253
Investment (I)					
<i>v</i> = 1	—	—	1.022	0.803	—
5	0.342	0.387	0.682	0.406	0.339
9	0.241	0.283	0.567	0.402	0.116
13	0.309	0.285	0.297	0.297	0.269
17	-0.059	-0.074	-0.038	-0.452	0.336
21	0.230	0.220	-0.387	-0.583	0.349
Central Government Expenditure (G)					
<i>v</i> = 1	—	—	1.416	1.282	—
5	0.844	0.910	1.512	1.221	-0.097
9	0.289	0.349	1.504	1.120	0.009
13	0.582	0.527	1.488	1.170	0.015
17	1.694	1.544	1.392	1.209	0.095
21	0.995	1.074	0.782	0.573	0.610
Gain ^{2c}					
<i>v</i> = 1	—	—	—	—	—
5	0.262	0.232	0.259	0.243	0.232
9	0.232	0.253	0.271	0.203	0.212
13	0.212	0.415	0.360	—	—

Table 4 continued

		Stock appreciation (SA)			Exports (X)		
v=1	1-365	1-159	3-362	—	1-042	0-864	3-452
	5	0-765	3-216	0-394	5	0-675	3-357
	9	0-525	2-843	0-240	9	0-345	3-323
	13	0-855	3-504	—0-164	13	—0-048	3-173
	17	0-667	3-667	0-142	17	—0-242	2-917
21	0-708	0-443	3-171	—0-041	21	—0-582	3-162
Corporate Profits (CP)							
v=1	0-570	0-407	3-510	—	1-410	1-258	3-549
	5	0-255	3-705	0-261	5	0-841	3-132
	9	0-116	2-904	0-382	9	0-476	3-172
	13	0-535	2-774	—0-419	13	0-084	3-305
	17	0-382	2-984	0-154	17	—0-365	0-449
21	—1-145	—0-442	3-071	0-526	21	—0-782	3-477
Factor cost adjustment (FCA)							
v=1	1-363	1-194	3-492	—	1-363	1-194	3-492
	5	0-815	3-139	0-549	5	0-539	3-139
	9	0-639	3-254	0-175	9	0-400	3-254
	13	0-548	3-115	0-090	13	0-266	3-115
	17	0-499	3-310	—0-010	17	0-276	3-310
21	0-374	0-092	3-117	0-124	21	0-092	3-117
Stockbuilding (SB)							
v=1	0-314	1-112	3-377	—	0-314	1-112	3-377
	5	0-894	3-378	0-420	5	0-692	3-378
	9	0-762	3-239	0-173	9	0-519	3-239
	13	0-115	3-553	0-554	13	—0-035	3-553
	17	—0-029	3-578	0-144	17	—0-173	3-578
21	—0-254	—0-428	3-473	0-225	21	—0-428	3-473

^a The 5% critical value for n , the sample size, = 45 is approximately 3.29 for $m = 4$ (as used here); if the sample value of K is less than this critical value we reject the null of normality. Critical values for $n > 50$ have not been tabulated; however, as the critical values increase with the sample size the least we can say is that a sample value below 3.34 ($m = 4$, $n = 50$) leads to rejection of the null.

^b $\text{Gain1} = \ln [(C_v - \mu_v)/(C_{v+4} - \mu_{v+4})]$ = successive difference between rows of column 1, subject to rounding.

^c Gain2 = successive difference between rows of column 2 subject to rounding.

square revision attributable to particular quarters. Quarter 1 again shows a higher average than the other quarters, so there does seem to be something of a 'quarter' effect in the revisions process.

Although pooling the components is of interest we need to go further and consider the individual components of GDPI and GDPE. However, as there are 5 components of the former and 7 of the latter and 44 quarters, some summary of the 'quarter' coefficients is required. Thus, Table 3 reports the average, by year and quarter, of the 'quarter' coefficients. Considering the GDPI components we see that the tendency for the 'quarter' coefficients to increase over time is a common pattern in the 5 components. Normalising the 'quarter' coefficients leads to a more stable pattern, but there are still some notable variations. For example, 1975 is associated with a particularly large mean square revision for corporate profits, gross trading surplus of public corporation and other income. The average across quarters shows that Q1 has the largest 'quarters' coefficient for 3 of the variables—corporate profits, stock appreciation and other income; with the Q1 coefficient second largest for the 2 remaining variables.

Amongst the GDPE components we find the downward trend in the normalised 'quarter' coefficients rather more pronounced for investment and exports than for the other variables—and there is a reverse tendency for FCA. No one quarter or year seems to have resulted in uniformly large mean square revisions for every variable. Years of note, associated with relatively large normalised coefficients are: 1973 for consumption; the period 1970 to 1974 for investment, after which three seems to have a notable reduction in the mean square revision attributable to the quarter effect; 1975 for government expenditure; and 1971 to 1973 for both exports and imports.

3.2 *Estimating the gain arising from later vintages of data*

We showed earlier that the gain, comparing two vintages, can be taken as the logarithmic difference in the 'vintage' coefficients if the two conditional distributions are normal and have zero means. In practice neither of these conditions is generally satisfied, and the reduction in uncertainty should be calculated directly by comparing the entropy of each conditional distribution.

The results in Table 4 are presented for $\delta=4$, corresponding to an annual pattern of revision. For the pooled variables this results in 220 and 308 observations for each vintage of GDPI and GDPE respectively, and 44 observations for each individual variable. Table 4 gives the following details. Column 1 is the entropy of the conditional distribution calculated using the variance of the empirical distribution assuming that the distribution is normal. Column 2 is the directly estimated entropy of the empirical distribution. The difference between columns 1 and 2 is, therefore, a broad indication of the extent to which the empirical distribution departs from normality. This idea is formalized in column 3 which reports Vasicek's 'K' statistic. Column 4 reports the gain calculated on the assumption of normality but without assuming that the conditional means are zero—see (7) above; and column 5 reports the gain calculated as the difference between the directly estimated entropies. These are called Gains 1 and Gain 2, respectively.

A visual impression of the empirical distributions is given in Figs 1 and 2, which show the histograms of the empirical (conditional) distributions of the pooled and individual components for $v=9$. Of these distributions the ones for the pooled components look closest to being normal; those for the individual components show distinct signs of non-normality.

We consider first the results for the GDPI components. Pooling these gives a distribution which does not seem too far from normal—except for $v=21$. Thus, we anticipate that Gain 1 and Gain 2 will be broadly similar except for the last gain and on the whole this is borne out. Note from Gain 2 that the uncertainty reduction does not decline

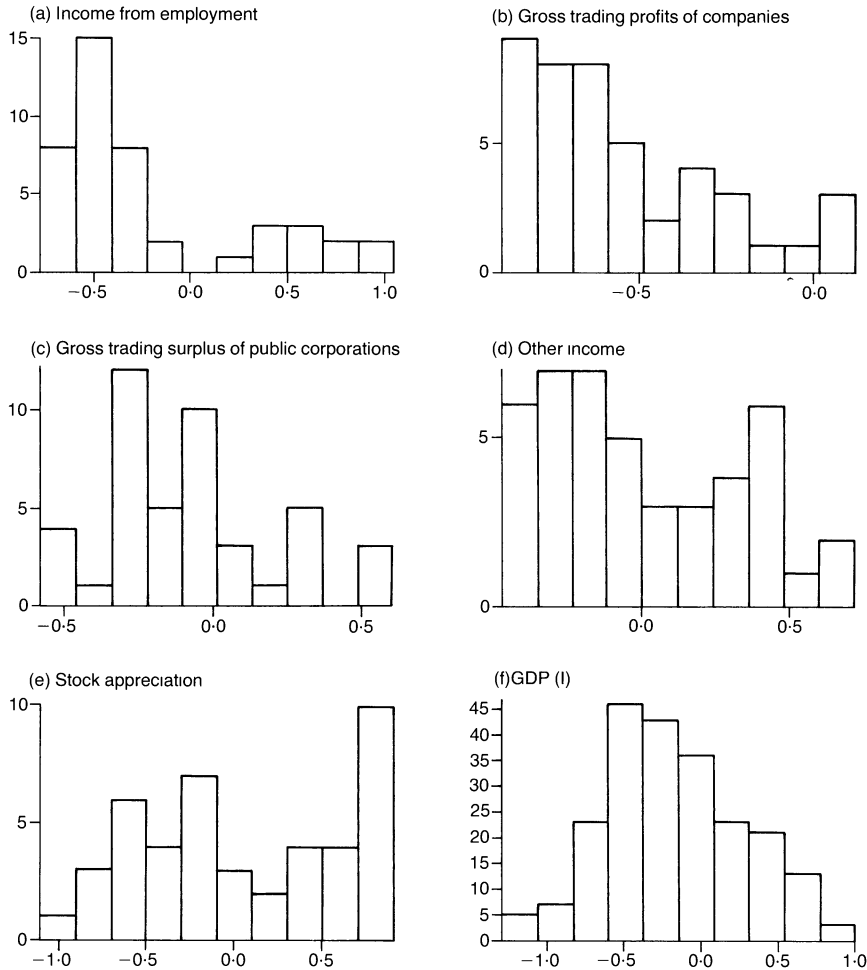


Fig. 1. Frequency distribution of the errors for components for GDP (I).

monotonically; vintage 5 shows a large gain relative to vintage 1, but we also find large gains from vintages 21 and 17. Income from employment is the largest component of GDPI. Here we find marked evidence of non-normality in most of the conditional distributions and, hence, we expect Gain 1 and Gain 2 to differ. Vintages 9 and 17 show the largest gains—see Gain 2—and vintage 21 seems to ‘overshoot’ in that the gain is *negative* for this vintage. The pattern for other income is one of similarly sized gains at each vintage except the 17th. Gross trading surplus of public corporations shows very large gains at $v=5$, 17 and 21. The largest gain for stock appreciation is for $v=5$, with some evidence of ‘overshooting’ at $v=13$; and the largest gain occurs at $v=21$. The evidence of non-normality is quite marked throughout the conditional distributions, with 22 of the 30 ‘K’ statistics indicating rejection of the null hypothesis at the 5% significance level.

Pooling the components for GDPE we find that the columns for Gain 1 and Gain 2 do not show any substantial difference, reflecting the closeness of columns 1 and 2 and hence of the empirical distributions to normality. The gains from vintages 5 to 17 are of a similar magnitude with a sizeable gain to come at vintage 21. Turning to the components we look first at consumption which is the largest component of GDPE. We find a large gain at $v=5$ and some ‘overshooting’ at $v=17$. The ‘K’ test indicates rejection of the null hypothesis at the 5% significance level for $v=9$. There is rather more evidence of non-normality for investment, with rejection of the null for $v=5$, 13 and 17. There is virtually

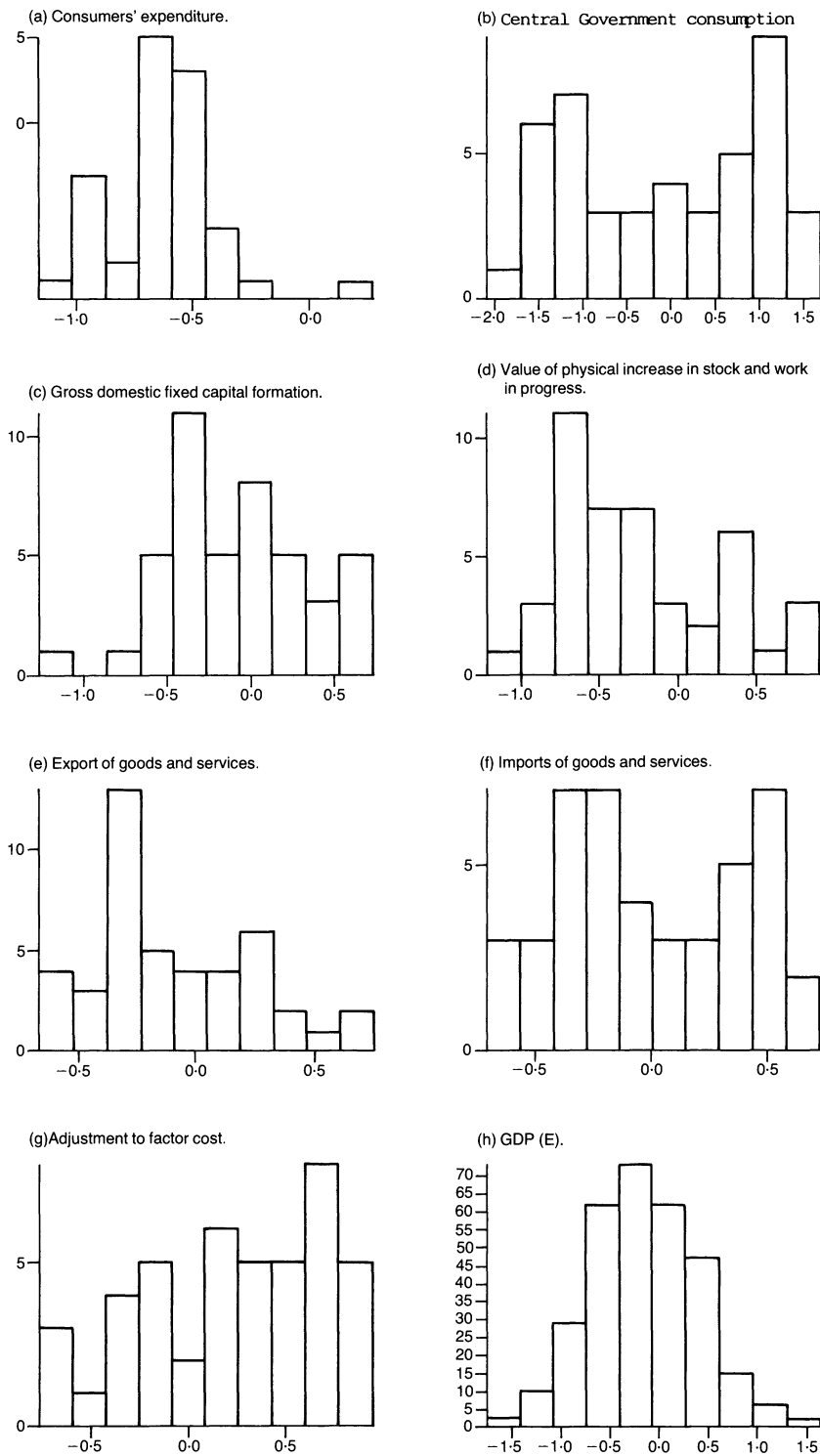


Fig. 2. Frequency distribution of the errors for components of GDP (E).

no change in uncertainty due to vintage 9, with largish gains for $v=5$, 13 and 17. The pattern for central government expenditure is in contrast to the other GDPE components. The estimated entropy of the conditional distributions shows very little change between vintages 1 and 17, the only real gain coming at $v=21$. The pattern for exports is one of gains throughout the revisions process, the largest gain coming at $v=13$. Turning to imports we find the largest gain at $v=5$, and then a number of gains of similar magnitude. The largest gain for factor cost adjustment is for $v=5$, with some 'overshooting' at $v=17$; gains for the other vintages are of a similar size. Finally, stock appreciation shows relatively large gains for $v=5$ and 13, whilst the gain for $v=21$ is larger than for $v=13$ or 17. Overall, there is evidence that about one half of the conditional distributions for Investment, Central Government Expenditure, Exports, Imports and Factor Cost Adjustments are not normal using the 5% critical values of the 'K' test. Whereas only one each of the distributions for Consumption and Stock Appreciation are non-normal using the 'K' test.

What is clear from these results is that we cannot say that uncertainty reductions decline as we use later vintages of data, neither can we ignore the later vintages of data as would be the case if the gains were close to zero for the later vintages. The evidence of non-normality of the conditional distributions suggests that the assumption of normality would give rise to misleading conclusions and hence direct estimation of the entropy is, in general, preferable.

4 Concluding remarks

This study has developed and applied a methodology for the analysis of different vintages of data. In particular, we have focused on the pattern of the revisions process for two data sets comprising the current price income and constant price expenditure components of GDP for the United Kingdom. These variables form the basis of much macro-economic modelling and it is, therefore, of some importance to know whether using, for example, the first rather than the last vintage of these series, so as to maximise the sample size, is likely to lead to a loss of information. Our developments have led to the wider applicability of a methodology initially developed by Theil. In particular, it can now be applied even if the underlying distributions are not normal; there is no need, for example to force the data through a transformation to try and obtain normality. Further, even if the hypothesis of normality is statistically acceptable, there is a need to adjust for non-constant and non-zero means in the conditional distributions. This is likely to be of practical importance both for data vintages and the evaluation of multi-stage forecasts.

Our empirical analysis shows that it is important to take account of the non-normality in the conditional distributions for the component variables. Over two-thirds of the distributions for the GDPI components and about half of those for the GDPE components lead to a rejection of the null hypothesis of normality. The information gain should, therefore, be calculated as the difference between the directly estimated entropies of the relevant conditional distributions.

The pattern of the information gains for our data sets shows that the trade-off between sample size and uncertainty reduction is, in general, likely to be a very real one. We found rather general evidence that there were gains associated with an annual revision process throughout the range of vintages. Thus, for example, the situation could not be characterised as one in which, say, just the first (annual) revision was sizeable compared with subsequent revisions. Hence, researchers using these data sets for empirical and policy analysis should be aware that their results need to be assessed for sensitivity with respect to the particular vintages being used. Not to do so would leave open the possibility that using unadjusted initial data and data of a substantially later vintage could lead to

different conclusions. If recommendations, either from empirical research or empirically based policy design, are sensitive to the vintage of data used then it would seem prudent to assess the prospect of modelling the relationship between different vintages. We hope in this study to have drawn attention to an important aspect of research and policy design which should have much higher profile than it seems presently to have.

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